

CSE11C Lab Assignments Manual

Delft University of Technology

Edition 2026-2027¹

¹Revision 20260902T0921

Acknowledgements

The CSE11C lab course assignments were originally developed by Sidney Cadot for the PowerPC architecture. The accompanying documents, two versions of the assignments and a tutorial on PowerPC assembler, were also written by Sidney. Jonne Zutt maintained these documents over the years after Sidney left the university. In 2004, the decision was made to change the target architecture of the lab course from the somewhat obscure PowerPC to the ubiquitous Intel x86 platform. The lab course environment had to change accordingly: the carefully tweaked commercial IDE/PowerPC emulator running on Microsoft Windows was abandoned in favour of the GNU assembler and debugger, which are available on nearly every Linux distribution. Because of these changes, the assignments and the accompanying reference material were rewritten by Denis de Leeuw Duarte.

In 2005 a new curriculum has been started, which transformed the old lab into a more elaborate project for Software Technology students, while Media and Knowledge Technology students only had to do a trimmed down version. The manual was modified to match the new requirements by Bas van der Doorn and Sander Koning.

In 2011 and 2012 the manual was further updated by Mihai Capotă and Alexandru Iosup and in 2013 and 2014 the manual received a refresh and expansion by Otto Visser. In 2014, the decision was made to switch from the x86 to the x86-64 architecture and the manual was edited by Elvan Kula. Maarten Sijm restructured the manual in 2019 and during the Corona (no, not the beer) summer of 2020 further restructuring and editing was done by Sára Juhošová, Taico Aerts and Otto Visser.

During the summer of 2022, this manual was split into two parts: the assembly manual, dealing with the theoretical concepts, and the assignments manual, providing information about the lab and its assignments. This restructuring was done by Alexandru Postu, with additional help from: Alexandra Marcu, Ana Băltărețu, Ana Cristiana Marcu, Daniel Peter, Rareș Toader and Yiğit Çolakoğlu.

Contents

Acknowledgements	i
1 Rules, Regulations and Etiquette	1
1.1 Teaching Assistants: what they can and cannot do	1
1.2 Verifying Your Work	1
1.3 Rules & Regulations	1
1.3.1 Correct Specifications	2
1.3.2 Functionally Correct Code	2
1.3.3 Algorithmically Correct Code	2
1.3.4 Properly Commented Source Code	2
1.3.5 Stack frames	3
1.3.6 Official calling conventions	3
1.4 Deadlines and Schedule	3
1.5 Anti-Fraud Policy	4
1.5.1 Statement on AI	4
2 Getting Started	5
2.1 Setting Up Your Environment	5
2.1.1 Direct GCC Support	5
2.1.2 WSL: Windows Subsystem for Linux	5
2.1.3 Virtual Machine	6
2.1.4 Docker	8
2.2 Editors & Syntax Highlighting	9
2.3 Building, Running and Debugging Programs	10
2.3.1 Building & Running	10
2.3.2 Debugging with GDB	11
2.3.3 Debugging with lldb	11
3 Mandatory Assignments	12
3.1 ASSIGNMENT 1: Powers	12
3.1.1 Assignment description	12
3.1.2 Required Reading	13
3.2 ASSIGNMENT 2: Recursion	14
3.2.1 Assignment description	14
3.2.2 Required Reading	14
3.3 ASSIGNMENT 3: Memory	15
3.3.1 Assignment description	15
3.3.2 Required Reading	16
4 Bonus Content	17
4.1 ASSIGNMENT 4: A Colourful Discovery (150 points)	17
4.1.1 Assignment Description	17
4.1.2 Required Reading	18
4.2 ASSIGNMENT 5: Valid Parentheses (250 points)	19
4.3 ASSIGNMENT 6: Simplified <code>printf</code> Function (50–500 points)	20
4.3.1 Required Reading	21
4.4 ASSIGNMENT 7: Implement “diff” in Assembly (400 points)	22

4.4.1	Required Reading	23
4.5	ASSIGNMENT 8: Hide Data in a Bitmap (500 points)	24
4.5.1	Required Reading	25
4.6	ASSIGNMENT 9: Branch Prediction (200–600 points)	26
4.6.1	Implementation	26
4.6.2	Testing your solution	27
4.6.3	Hints	27
4.7	ASSIGNMENT 10: Brainfuck (400 points)	28
4.7.1	Assignment description	28
4.7.2	Required Reading	28
4.8	ASSIGNMENT 11: Implement a Game (750 points)	29
4.8.1	Assignment description	29
4.8.2	Hints about Solving this Assignment	30

Chapter 1

Rules, Regulations and Etiquette

This chapter elaborates on aspects that you should consider during the labs. The role of Teaching Assistants (TAs) will be detailed in Section 1. Section 2 will elaborate on the submission process. Section 3 will briefly explain the necessary in order to get a solution approved. Lastly, sections 4 and 5 highlight the importance of deadlines and of not committing fraud.

1.1 Teaching Assistants: what they can and cannot do

Teaching assistants will be present during lab course hours to offer you advice and the possibility to have your work reviewed. Please keep in mind that they are *not* there to deal with faulty lab course equipment, problems with your laptop and software, problems with your login account, disputes about deadlines and lab course rules, possible special exceptions (e.g., due to illness on your part) or problems with lab course enrolment. We kindly ask you to bypass the teaching assistants in such matters and to go straight to the proper authorities (see table below).

Login accounts:	Service point
Deadlines, rules, exceptions:	Course Coordinators (co-cs-ewi@tudelft.nl)
Enrolment:	Academic Counsellors (ac-bsc-ewi@tudelft.nl)

1.2 Verifying Your Work

In order for you to get the credit that you passed a lab exercise, you need to have your work approved by a teaching assistant. For this, you will have to submit your work to Submit¹ and then enqueue to the digital Queue². Please enable notifications on this page. Once one of our assistants picks up your request, you will receive a notification. Once that happens, kindly wait for the TA to arrive at your spot. If your request has been picked up by an assistant but you are not found in the described spot, your request will be denied and you will have to enqueue again.

1.3 Rules & Regulations

The work you hand in during the lab is subject to certain rules and quality guidelines, which will be explained in this chapter. Your work will only be considered if it fully complies with those. These rules effectively apply to *all* lab courses, but to avoid any confusion, they are stated here explicitly for this course.

1. You need to deliver *correct specifications*.
2. Your finished code needs to be *functionally correct*.
3. Your code needs to be *algorithmically correct*.

¹<https://submit.tudelft.nl>

²<https://queue.tudelft.nl>

4. Your code needs to be *properly commented*.
5. Your code uses stack frames for all subroutines.
6. Your code uses the *official calling conventions*.

We stress that your work will not be considered fit for approval until you meet all of these criteria. In other words, if your assignment submission fails to meet any of the aforementioned points, you will **not** pass the assignment. The details on what we consider to be *correct* in this context will be defined in the subsequent paragraphs.

1.3.1 Correct Specifications

If an assignment so asks, you will need to hand in a *correct specification* for the specified part of the exercise. Specifications must be written in *pseudocode*. Pseudocode is code that resembles actual high level (non-assembler) program code closely. It is called pseudocode because it does not necessarily have to be real, compilable code. A good, clear example of pseudocode is provided in section 2.1 of the Assembly manual. We expect your pseudocode to be similar in clarity, quality and level of detail. Specifically, this means that your pseudocode must have:

- proper comments
- a good, clean layout
- a simple, clearly understandable structure

We expect that you have your specifications checked *before* you start on your implementation work. It is highly likely that the teaching assistants will ask you to make changes to your specifications, so we strongly advise you not to start programming before having your specs approved. If you choose to ignore this advice, we do not accept responsibility for any wasted effort on your part.

1.3.2 Functionally Correct Code

The source code that you hand in should be functionally correct. This is implied by the following:

- The source code must compile without errors or warnings.
- The program must function as specified in the assignment.
- The program must run to completion without runtime errors.

1.3.3 Algorithmically Correct Code

The programs that you hand in should be *algorithmically correct*. The correct algorithm is, by definition, the algorithm that was conveyed to you by the teaching assistants during the approval of your specifications. This implies that a program cannot be algorithmically correct if you did not have your specifications approved by a teaching assistant. We explicitly state here that you are *not* free to implement the desired functionality at your personal leisure and that solutions which are merely functionally correct are not sufficient for approval.

1.3.4 Properly Commented Source Code

All source code, including pseudocode, should be properly commented. Commenting source code is not an exact science, but at the least we expect that you adhere to the following guidelines:

Natural language comments should be written in a natural human language, which is preferably in British English. Pseudocode and overly complicated mathematical notations are not accepted as proper comments.

Continuity the comments in the body of your code should “tell the story” of the code in a concise, clear manner. If you strip away the accompanying source code, your comments should make up a compact description of the algorithm in acceptable prose. Do place your comments with care and avoid overly lengthy comments.

Subroutines Each subroutine should be accompanied by a clear description of its function. A description should also be given of the meaning and type of the arguments and the return value.

Layout Part of the readability of source code comes from good layout. We expect you to be precise and, most importantly, consistent in your style.

You are allowed to develop your own style of commenting and layout, but we demand that you be consistent and precise. Sloppy comments or disturbing layouts will not be accepted.

1.3.5 Stack frames

Assembly is not a high-level programming language. Consequently, more thought has to be put into the behaviour of the stack. For instance, your code must always clean the stack after every function call.

For an explanation of everything that you must consider, kindly refer to section 2.5 of the Assembly manual.

1.3.6 Official calling conventions

This relates to writing subroutines (which you will be required to do for all assignments). You are expected to respect the official calling conventions at all times. That is to say, registers should be used in a specific order and for specific scenarios. When you can no longer use just registers alone for your program, you should start using the stack.

For additional information, consult section 2.6 of the Assembly manual.

1.4 Deadlines and Schedule

In addition to the rules listed in this chapter, there may be hard deadlines. Please check the Brightspace pages for the most up-to-date information regarding those. Please note that teaching assistants are not allowed to consider any work after the expiration of a deadline. Any complaints should thus be addressed to the course coordinators.

The following table gives an overview of the **recommended** schedule for the lab. For a more detailed deadlines overview, kindly check the Brightspace calendar: <https://brightspace.tudelft.nl/d21/1e/calendar/860447>.

Week 1:	Find a lab partner, read the lab manuals and set up the environment
Week 2:	Assignment 1
Week 3:	Assignment 2
Week 4:	Assignment 3 and sign off the mandatory assignments
Week 5:	Midterm Exam
Week 6:	Bonus assignments (actual deadline for mandatory assignments)
Week 7:	Bonus assignments (actual deadline for bonus assignments and game specification)
Week 8:	Review materials (actual deadline for game)
Week 9:	Review materials
Week 10:	Endterm Exam

Note: We recommend signing off assignments as early as you are able to. There are limited sign-off slots available, and waiting until the deadline can mean that there are no longer enough slots for you to get (all) your assignment(s) signed off. Sign-off slots are provided on a first-come, first-served basis. Over the past years, the number of students submitting bonus assignments has significantly increased and labs have been full starting from the middle point onwards. If you implement a bonus assignment late, **we do not guarantee that you can also book a sign-off slot.**

You will *need* your time for this lab course, it is not easy. Many students underestimate the lab, do not start when they should and find out that they cannot complete it anymore. Do not let this happen to you and make sure you visit every session, so you can talk about your questions to the assistants. Outside lab hours, we encourage you to use <https://answers.ewi.tudelft.nl>.

1.5 Anti-Fraud Policy

Our anti-fraud policy is very simple: zero-tolerance, within the limits set by TU Delft. We will pursue each case of potential fraud, and will use the means provided by TU Delft to punish (attempts to) fraud. The following are some of the cases that are considered fraud:

- Sending your code to other groups. The motivation of “I sent it for them to find some inspiration” does not work.
- Copying somebody else’s code. Changing the names of variables in someone else’s code and submitting the results is still considered fraud.
- Receiving help from someone, when the help amounts to letting that someone write your code.
- Renting the services of a programmer, for example from **Rent-a-Coder.ro**, to solve the assignments for you.

1.5.1 Statement on AI

In this course, the use of generative AI is strictly forbidden for any type of use unless specifically stated otherwise in the assignment. This includes programming exercises, self studies, and all other work you submit. You are expected to implement your own code and brainstorm your own ideas. Teaching assistants are instructed to pay attention to potentially AI-generated code or answers. Using AI for assignments is considered fraud.

Chapter 2

Getting Started

This chapter contains all the information you will need to get started on this lab, including how to set up your environment, tips on which applications to use, and instructions on how to compile, run, and debug your code.

2.1 Setting Up Your Environment

There are multiple ways in which you can set up your environment, depending on your operating system. The option we recommend for your system is highlighted in bold.

	Windows		macOS	Linux
	Windows 10/11	Other		
Direct GCC Support	No	No	No	Yes
Windows Subsystem for Linux	Yes	No	No	No
Virtual Machine	Yes	Yes	Yes (Intel CPUs only)	Yes
Docker	(Yes)	(Yes)	Yes (Apple Silicon)	(Yes)

2.1.1 Direct GCC Support

Supported systems: Linux

If your system is listed under direct GCC support, you can simply install the required software on your system as explained below.

Linux

Depending on your Linux distribution, the installation will be different. Use the package manager of your Linux distribution to install the `gcc` and `gdb` packages. We support GCC v4 or above, but we will use v12 when checking your work. For example, for Debian-based distributions like Ubuntu, simply write the following command in your terminal:

```
sudo apt install build-essential gdb -y
```

If you are running another Linux distribution, we will assume you know what you are doing and will be able to install `gcc` and `gdb` yourself (we recommend the package `build-essential`).

2.1.2 WSL: Windows Subsystem for Linux

Supported systems: Windows 10 and 11

Windows includes the option to install and run Linux inside of Windows. It is simple to install and set up. *Please note, however, that if you want a graphical interface for the “Create a Game” assembly bonus assignment, you have to use a virtual machine for it to display on Windows.*

Installation

Follow the instructions on <https://docs.microsoft.com/en-us/windows/wsl/install> (i.e. the part under “Install WSL command”).

Issues? Then instead follow the instructions on <https://docs.microsoft.com/en-us/windows/wsl/install-manual>. When instructed, install the “Ubuntu” app from the store (the one without version number).

You can now launch Ubuntu from your start menu. The first time you launch it, you will be asked to set a username and password for your Linux account (be sure to remember these or write them down).

Finally, run the following command to install the necessary build tools:

```
sudo apt update && sudo apt upgrade -y && sudo apt install build-essential gdb -y
```

For more instructions and troubleshooting, see <https://docs.microsoft.com/en-us/windows/wsl/troubleshooting>.

Terminal

The default terminal that Windows uses for WSL is suboptimal and lacks support for features like blinking (which you need for bonus assignment 4). Instead, we recommend using the **Windows Terminal** (no, not the terminal that Windows comes with, but a program Microsoft made and named like this. We didn’t invent the name either). You can install it from the Microsoft Store: <https://www.microsoft.com/store/productId/9N0DX20HK701>

Once installed and launched, you can change the settings to use Ubuntu by default by clicking on the down arrow in the menu bar.

Finding files

When using WSL, we recommend putting your files in a location in Windows, e.g. somewhere in your Documents folder. You can access this location in WSL by using the `cd` command to navigate to the correct folder, though you will have to translate your Windows path to a Linux path. For example, if your files are at `C:\Users\Student\Documents\CO\Lab`, you should use the following command to navigate to it in WSL (case-sensitive!):

```
cd /mnt/c/Users/Student/Documents/CO/Lab
```

Should you want to access/edit the files created in a WSL-only location from Windows, you can do the following. In your Ubuntu terminal, enter `explorer.exe .` (with the dot!). This will open the current folder in Windows Explorer. Please note that this location of `\\wsl$\Ubuntu\...` is only accessible while WSL is running.

If this doesn’t work, open Windows Explorer and type the path `\\wsl$\Ubuntu\home`.

Visual Studio Code

Visual Studio Code has an extension for easy integration with WSL. You can find it at <https://marketplace.visualstudio.com/items?itemName=ms-vscode-remote.remote-wsl>.

2.1.3 Virtual Machine

Supported systems: All but M-series Macs

With a virtual machine you can run a virtual computer as a program on your computer. We provide you with a virtual machine (VM) with Debian and the necessary tools already installed. We also pre-installed useful editors and their assembly syntax highlighting support (see Section 2.2). You can run this on your own machine using VirtualBox.

Installing and Starting

1. Download the virtual machine from Brightspace (Content → Resources → Lab → Virtual Machine).
2. Download VirtualBox from <https://www.virtualbox.org/wiki/Downloads> and install it.
3. Open Oracle VirtualBox.
4. Click the “Import” button.
5. Select the virtual machine you downloaded in step 1 and click Next.
6. Leave all settings at the default settings and click Import.

The virtual machine is now ready to be used. Click the “Start” button (green arrow) to start it. To log in use the following credentials:

username: student
password: pwd

Troubleshooting

The virtual machine is very slow.

You can increase the number of CPU cores available to the virtual machine which might alleviate this problem. Select the virtual machine, click Settings → System → Processor and increase the number of processors. Ensure that the amount selected falls within the green bar displayed by VirtualBox.

Not enough memory.

If you don't have enough memory to start the virtual machine, first try closing all unnecessary programs on your computer. You can adjust the amount of memory made available to the virtual machine as follows: Select the virtual machine, click Settings → System. Now set an appropriate amount of RAM (minimum 1024 MB). Ensure that the amount selected falls within the green bar displayed by VirtualBox.

(Windows) Hangs on black screen during boot.

If you are using Windows and your virtual machine keeps getting stuck on a black screen while starting up, make sure that Hyper-V is enabled on your machine. To enable it (or check if it is enabled), perform the following steps:

- Press start.
- Search for “Turn Windows features on or off”.
- Find “Hyper-V” in the list.
- Ensure the checkbox of Hyper-V is fully checked (a check mark, not a square).
- Press “OK”.
- Let the process finish and restart your PC when prompted.
- Try the virtual machine again.

VT-X/AMD-V is not available.

If the VM refuses to boot and gives an error like “VT-X/AMD-V is not available”, please make sure that the virtualization is enabled in your BIOS. For example, the HP Workstations from the TU Delft Laptop Project do not have this setting turned on by default. Here's a list of steps on how to turn virtualization on:

- Go to your BIOS settings. This can usually be done by pressing F12 while booting (although it varies based on machine - we encourage you to google what works for yours). To enter your BIOS settings from Windows, the following approach **might** work, based on your machine:
 - Press start.
 - Click the power icon.
 - Hold shift while clicking “Restart”.
 - Wait for a bit until the Windows UEFI settings appear.
 - Click “Troubleshoot”.

- Click “Advanced options”.
- Click “UEFI Firmware settings”. This will reboot your PC into the BIOS settings.
- Inside the BIOS Settings, go to the “Advanced” menu, then to “System”, and enable “Virtualization options”.
- Make sure to “Save and Exit”, and not just “Exit” the BIOS Settings.
- Once your PC has rebooted, you should be able to run the virtual machine in VirtualBox. If not, please ask the lab assistants for help.

For other laptops, the BIOS setting might have a different name or be under a different category. Try googling for “enable virtualization on <model of your laptop>” to find instructions specific to your laptop.

2.1.4 Docker

Supported systems: All, including M-series

On an M-series chip (Apple Silicon) MacBook you cannot use our virtual machine. Instead, you can use Docker.

Although this approach works for all systems, we suggest you try the previously mentioned ones. That is because this approach leads to limited functionality when it comes to debugging. This can lead to a harder time completing the assignments, especially when it comes to the more difficult ones.

1. Download Docker, from:

- <https://docs.docker.com/desktop/install/windows-install/> (for Windows)
- <https://docs.docker.com/desktop/mac/apple-silicon/> (for Apple Silicon)
- <https://docs.docker.com/desktop/install/mac-install/> (for previous Mac versions)

2. Install and open it

3. `docker run --name <insert_name_here> -it \`
`-v ~/asm_projects:/root/asm \`
`--cap-add=SYS_PTRACE \`
`--security-opt seccomp=unconfined \`
`--platform=linux/amd64 \`
`debian:latest bash`

Now you can use these steps to run a previously created container:

1. Start the container

- (a) From Docker directly
- (b) In terminal: `docker start <container_id_or_name>`

2. Run the container:

```
docker exec -it <container_id_or_name> /bin/bash
```

3. Now you are inside an OS in the terminal! :)

Note: if it is your first time running the container, you have to install and update the tool-chain:

```
apt update && apt install build-essential lldb vim -y
```

Inside the container, you can run gcc and gdb as explained later in the manual. (unavailable on M-series MacBooks, but lldb can be used as an alternative) You can use vim or nano as editor inside the container. We recommend to use your preferred editor in macOS, and just copy pasting your code into a file in the container (e.g. in nano: `nano myfile.s`).

2.2 Editors & Syntax Highlighting

You are free to write your code in a text editor of your choice. There are various text editors which support syntax highlighting of x86 assembly which can be helpful for the assignments. We recommend a text editor such as Sublime Text (<https://www.sublimetext.com/>), Visual Studio Code (<https://code.visualstudio.com/>), gedit, vim or emacs. We will explain how to add/enable syntax highlighting for the editors that we recommend.

Sublime Text

You can add syntax support for x86 assembly as follows:

1. Go to Tools → “Install Package Control...” and wait for the installation to complete
2. Go to Tools → “Command Palette...” and enter “Package Control”
3. Select “Package Control: Install Package” (See Figure 2.1)
4. Enter x86, click on GAS-x86 (See Figure 2.2)

To enable the highlighting for a file, select View → Syntax → GAS/AT&T x86/x64. You can also set it as the default for all files of the same extension by going to View → Syntax → Open all with current extension as... and then selecting GAS/AT&T x86/x64.

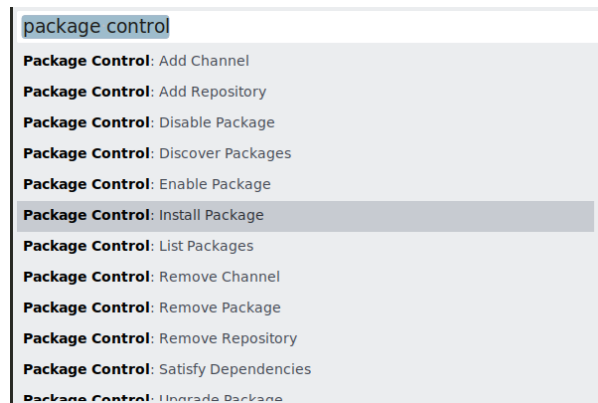


Figure 2.1: Installing a package in Sublime Text

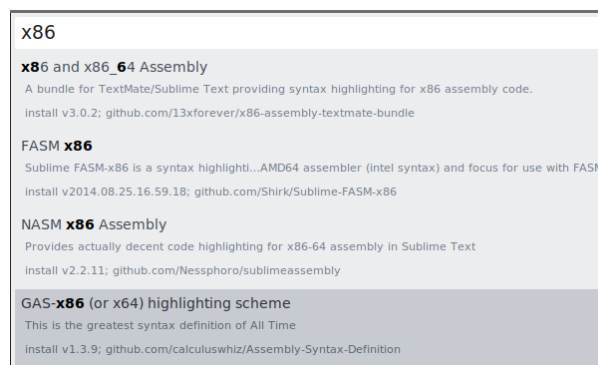


Figure 2.2: Installing x86 Syntax Highlighting for Sublime Text

Visual Studio Code

Install GNU Assembler Language Support as follows:

1. Launch VS Code Quick Open (**Ctrl** + **P**)
2. Paste `ext install basdp.language-gas-x86`

3. Press enter

Save your files with the `.s` extension and VS Code will automatically enable the syntax highlighting.

gedit

To install x86 assembly syntax highlighting for gedit, open a terminal and paste the following commands:

```
wget https://raw.githubusercontent.com/calculuswhiz/gedit-GAS-x86_64-highlighter/master/GAS_x86-64.lang
sudo cp GAS_x86-64.lang /usr/share/gtksourceview-3.0/language-specs/
```

Save your file with the `.s` extension and gedit will automatically enable the syntax highlighting. Otherwise, go to View, Highlight Mode..., and select “GAS (x86-64)”.

VIM

Ensure that `git` is installed, then execute the following:

```
git clone https://github.com/calculuswhiz/vim-GAS-x86_64-highlighter.git
cd vim-GAS-x86_64-highlighter
sh install.sh
```

Save your file with the `.s` extension and vim will automatically enable the syntax highlighting.

Emacs

Emacs has an `asm-mode` which adds some support for writing assembly. There is a GAS mode for Emacs as well, which you can download at <https://github.com/Taeir/emacs-gas/blob/master/gas-mode.el>.

2.3 Building, Running and Debugging Programs

For your programs to be able to run, you need to compile them into an executable file. The following section contains instructions on how to do exactly this.

2.3.1 Building & Running

The short story: open a terminal window, navigate to the directory that contains your sources and run the following commands:

```
gcc -no-pie -o nameofyourprogram nameofyoursource.s

./nameofyourprogram
```

where `nameofyourprogram` is the name you want to give your executable file and `nameofyoursource.s` is the file in which you wrote your piece of code.

The longer story: in order to create an executable program out of your assembly source code, you need to *assemble* it using a tool called `gas`, the GNU assembler. This creates a so-called *object file*¹. Since your actual program may consist of subroutines that are defined in different object files and libraries, the resulting object file(s) needs to be linked into an executable using the tool `ld`, the linker. Unfortunately, it is also necessary to include a host of other files in the linking process, such as the standard C library and the C runtime environment, to produce a proper Linux executable. The exact files to link may differ from one Linux distribution to another and the list may become rather long, which is why we do what every sane person does: we swallow our pride and cheat by simply using `gcc`, the GNU compiler collection, to call `gas` and `ld` on our behalf. If you are curious as to how bad the actual calls look, try using `gcc` in verbose mode with the `-v` flag. You will see that we did not succumb without a battle:

```
gcc -v -no-pie -o nameofyourprogram nameofyoursource.s
```

¹This has nothing to do with objects in the Object-Oriented Programming sense.

The `-no-pie` flag has little to do with pie; PIE stands for Position Independent Executable and is an effort to make binaries less predictable and hacking harder. Unfortunately, it would make the assignments a bit harder as well, so you will have to provide `-no-pie` for now to disable it. More information can be found on the Internet ².

2.3.2 Debugging with GDB

When your program gives unexpected output or a **Segmentation Fault**, one way to debug your program is to carefully read through your assembly code and try to find what's wrong. Another (less painful) way is to use GDB.

In order to use GDB, pass the `-g` flag to the compiler as shown in the following command:

```
gcc -no-pie -g -o nameofyourprogram nameofyoursource.s
```

This will tell `gcc` to compile your program with debug flags enabled. In order to start the debugger, type in the following command:

```
gdb ./nameofyourprogram
```

You will enter a `gdb` shell, where you can use some basic commands to instruct the debugger. For example, `break myfile.s:10` will set a breakpoint in the file 'myfile.s' on line 10³ and `run` will run your program. Once execution reaches your breakpoint, you take control over your program. Use `next` or `step` to execute the next line of code, the difference being that the former will execute an entire subroutine call as one line, while the latter will step *into* the called subroutine. If you stepped into a subroutine but already saw what you wanted to, you can use `finish` to execute the rest and step out of it. To resume execution, use `continue`. We recommend that you use `layout regs` before running your program, which will allow you to see the registers' values while you step through the code. Finally, you can `quit`.

Many other commands can be found (for example) in this cheatsheet⁴.

2.3.3 Debugging with lldb

While debugging with `lldb` has a slightly steeper learning curve than `gdb`, it is the most reliable way to debug x86 assembly natively on Apple Silicon (M-series) Macs.

Assuming you have successfully compiled `nameofyourprogram.s` into an executable named `nameofyourprogram` (as presented in 2.3.2), you can start a debugging session by running the following commands in your terminal:

1. `ROSETTA_DEBUGSERVER_PORT=1234 ./nameofyourprogram &`
2. `lldb ./nameofyourprogram`
3. `gdb-remote 1234` (*run this inside the lldb prompt*)

In this shell debugger, you can write many helpful commands, such as: `b main` to set a breakpoint at `main`, or `register read rax` to read the `%rax` register. Many other commands can be found (for example) in this cheatsheet⁵.

²https://en.wikipedia.org/wiki/Position-independent_code#Position-independent_executables

³You can leave out the file name (e.g. `break 10`) when you just have one assembly file.

⁴Cheatsheet for GDB: <https://www.cs.princeton.edu/courses/archive/fall16/cos432/hw2/gdb-refcard.pdf>

⁵Cheatsheet for lldb: <https://firefly.com/lldb-cheatsheet/>

Chapter 3

Mandatory Assignments

This chapter contains the description of all mandatory assignments. There is a separate module on Submit for each separate assignment, to which you should upload your code. Once you have uploaded all mandatory assignments, and all tests for each assignment pass (they automatically run when uploaded), a Teaching Assistant can sign off all three assignments in one go. Before trying to get your solutions approved by a Teaching Assistant, make sure all tests for all assignments pass, and read 1.3.

3.1 ASSIGNMENT 1: Powers

This section will detail the first mandatory assignment that you need to do. For the assignment description, refer to 3.1.1. A list of the theoretical concepts that you require to do this assignment can be found under 3.1.2.

3.1.1 Assignment description

This first assignment consists of two parts. Part A will get you started with assembly and how to build and run your own assembly programs. In part B, you will write your first *subroutine* with I/O and some basic programming constructs. You will only need to hand in part B, but you can ask an assistant (not make a submission!) whether you are on the right track after finishing part A.

Part A: Getting Started

In this assignment you will be asked to write your first assembly program. You will have to use the knowledge you acquired from the example program (found under section 2.1 of the assembly manual) in order to complete this task, so make sure you have a thorough understanding of it. Remember that you can always ask the lab course assistants for help. For this program, you will not have to write any specifications, since there is no significant algorithmic complexity involved. However, you are of course required to write proper comments.

Exercises:

1. Create a new text file, called “power.s”.
2. Implement a simple `main` routine that exits the program immediately with the proper exit code and without crashing.
3. Build your program and run it.
4. Alter your `main` routine in such a way that it prints a message containing your names, netIDs and the name of the assignment on the terminal.

You should not need more than one call to `printf` to display your message. After completing the rest of the exercises in part B, you will need to have the source code of this program checked by the teaching assistants, so make sure you keep all your files in order.

Part B: Your First Assembly Algorithm

Now that you have successfully run your first assembly program, it is time to write a more complex program. In this assignment you will write a subroutine that takes several input parameters and returns a computed value.

Exercises:

1. The following partial specification of the `pow` subroutine is given:

```
/**
 * The pow subroutine calculates powers
 * of non-negative bases and exponents.
 *
 * Arguments:
 *
 *     base — the exponential base
 *     exp  — the exponent
 *
 * Return value: 'base' raised to the power of 'exp'.
 */
int pow(int base, int exp) {
    int total = 1;

    // ...

    return total;
}
```

Complete the specification of the `pow` subroutine. You should only use looping constructs and simple arithmetic operations to compute the total.

It is not required to sign off this specification, but you can ask a teaching assistant to check that it is correct. When you have questions about this assignment, the teaching assistant will ask for your specification.

2. Create a new subroutine called `pow`, which will be the implementation of your `pow` subroutine. Make sure the subroutine takes 2 inputs - first one should be the base and second one should be the exponent, and has 1 output - the base raised to the exponent.
3. Alter your `main` routine in such a way that it asks the user for a non-negative base and exponent. The first input should be the base and the second input should be the exponent.
4. Alter your `main` routine in such a way that it calls `pow` with the numbers it reads, and prints the result of `pow` on the terminal.

3.1.2 Required Reading

1. You **must** respect the calling conventions. To read about them, check subsection 2.6.1 of the assembly manual.
2. To complete this assignment, you will need to use the **`printf`** subroutine. You may learn how to do so in subsection 2.7.1 of the assembly manual.
3. Section 2.3.1 of this manual explains the commands that you will need to enter on your shell in order to build and run your program.
4. To code the algorithm, you might find section 2.9 of the assembly manual useful. It describes programming constructs such as if statements and while loops.

3.2 ASSIGNMENT 2: Recursion

This section will detail the second mandatory assignment that you need to do. For the assignment description, refer to 3.2.1. A list of the theoretical concepts that you require to do this assignment can be found under 3.2.2.

3.2.1 Assignment description

By now, you should have a fairly thorough understanding of the stack mechanism and of its uses (e.g., storing local subroutine variables). In this assignment you are going to write a recursive subroutine that calculates the factorial of a number (“ $n!$ ”). This subroutine will be about 14 instructions in length when it is finished, but writing it will be fairly difficult. Do not be discouraged if it takes you a few hours to get it working.

Exercises:

1. Copy your “power” program to a new file called “factorials”. Create a new subroutine called **factorial**. This new subroutine should take one parameter, n , and for now it should simply do nothing and return n in the RAX register. Alter your **main** routine in such a way that it calls **factorial** with the number it reads, instead of calling **power**. **main** should print the result of **factorial** on the terminal.
2. Write a pseudocode specification of your **factorial** subroutine. The subroutine accepts a non-negative parameter n and it should return $n!$. Make sure your algorithm is **recursive**. It should not need to be more than a few lines of pseudocode.
It is not required to sign off this specification, but you can ask a teaching assistant to check that it is correct. When you have questions about this assignment, the teaching assistant will ask for your specification.
3. Implement your **factorial** routine. Test your program thoroughly.

3.2.2 Required Reading

1. This assignment requires you to write a recursive subroutine. For this, check subsection 2.6.2 of the assembly manual.

3.3 ASSIGNMENT 3: Memory

This section will detail the third mandatory assignment that you need to do. For the assignment description, refer to 3.3.1. A list of the theoretical concepts that you require to do this assignment can be found under 3.3.2.

3.3.1 Assignment description

Hundreds of years ago, archeologists discovered a treasure chest full of ancient scripts and drawings. Sadly, the puny minds of the current civilisation could not comprehend the contents of these precious documents and after decades of searching for a way to unlock their mysteries, people were forced to admit defeat and locked them away until someone worthy came along and could decode them.

Then, a few days ago, a strange machine was discovered. Our knowledge of technology is yet too limited to be able to get it working, but the group of scientists working on it managed to glean how the machine decodes these artifacts.

Now, they have tasked you with recreating it in the primitive ways of our technology. They have found that the artifacts are encoded in 8-byte memory blocks. The bytes in a memory block signify the following (from highest to lowest):

Byte 1 - 2	Unknown - this is still being worked on, but they have already determined that it is not crucial knowledge for decoding the messages.
Byte 3 - 6	The next memory block to visit.
Byte 7	The amount of times that character should be printed.
Byte 8	The ASCII ¹ character which should be printed.

So executed on the following memory blocks (where B = 00000000):

0	B B	B B B 00001000	00000001	01001000	?	8	1	H
1	B B	B B B 00001010	00000001	01010111	?	10	1	W
2	B B	B B B 00000111	00000001	01101100	?	7	1	l
3	B B	B B B 00000001	00000001	00100000	?	1	1	_
4	B B	B B B 00000101	00000010	01101100	?	5	2	l
5	B B	B B B 00000011	00000001	01101111	?	3	1	o
6	B B	B B B 00000000	00000001	00100001	?	0	1	!
7	B B	B B B 00000110	00000001	01100100	?	6	1	d
8	B B	B B B 00000100	00000001	01100101	?	4	1	e
9	B B	B B B 00000010	00000001	01110010	?	2	1	r
10	B B	B B B 00001001	00000001	01101111	?	9	1	o

your program should do the following:

1. Start at memory block 0.
2. Print the character **H** once and jump to memory block 8.
3. Print the character **e** once and jump to memory block 4.
4. Print the character **l** **twice**, and jump to memory block 5.
5. ...
6. Print the character **!** and terminate.

Which would result in the famous **Hello World!**

¹<http://www.asciitable.com/>

Exercises:

1. Download the files for this assignment from Brightspace.
The file you will be writing your code in is “decoder.s”. It currently includes the message from “helloWorld.s” (the `.include ‘helloWorld.s’` line does this). If you want to change the input message, simply change this line to include the file you want.
2. Write a pseudocode specification of your `decode` subroutine. The subroutine accepts the address of the message as its first parameter and has no return value. The following is an outline for your specification:

```
/**
 * The decode subroutine decodes the messages.
 *
 * Arguments:
 *     address — the address of the message
 *                in memory
 *
 * Return value: none
 */
void decode(int address) {
    // ...
}
```

You may also write helper subroutines which will be called from `decode`.

It is not required to sign off this specification but you can ask a teaching assistant to check that it is correct. When you have questions about this assignment, the teaching assistant will ask for your specification.

3. Implement your `decode` subroutine. Test your program thoroughly. Provided are test files “abc_sorted.s”, “helloWorld.s”, and “final.s”. We recommend starting with “abc_sorted.s”. This should print two lines: 0 through 9 on the first line and a through z on the second line. The memory is sorted, so this should work by each time just taking the next memory block. For Hello World you will need to extract the index of the next memory block. As for the contents of Final: the archeologists have not been able to figure this out yet. Can you?

This exercise wraps up the basic assembler programming assignments. You should go and have your code checked by a lab course assistant. Well done!

3.3.2 Required Reading

1. This assignment requires you to read the message from memory. You can learn to do so in subsection 2.3.2 of the assembly manual.
2. For this assignment, you need to be familiar with bit shifting. For this, check section 2.10 of the assembly manual.

Chapter 4

Bonus Content

This chapter contains only bonus assignments. For each assignment, you can receive extra points towards your final grade (the amount is always listed with the assignment). An overview of the bonus points is shown below:

Num	Assignment	Points
A4	Colourful Discovery	150 points
A5	Parentheses	250 points
A6	Printf	50 – 500 points
A7	Diff	400 points
A8	Bitmap	500 points
A9	Branch Prediction	200 – 600 points
A10	Brainfuck	400 + potentially extra points
A11	Game	750 points + potentially extra points

You have the freedom to select any assignment you prefer, but keep in mind that your points are capped at 2,000 (not including the extra points from Brainfuck / Game). Make sure you choose wisely!

4.1 ASSIGNMENT 4: A Colourful Discovery (150 points)

This section will detail the first bonus assignment that you can do. **It builds upon the work you did in Assignment 3 and as such can only be done with the same group you had for Assignment 3!** A list of the theoretical concepts that you require to do this assignment can be found under 4.1.2.

4.1.1 Assignment Description

Good news!

The team of scientists has figured out the last two bytes of our mysterious messages from the previous assignment: colours. So simple! The first byte is supposedly the colour of the background and the second one is the colour the character should be (foreground). Both of them are in the format of the ANSI 8-bit escape codes.

So returning to our example from section 3.3.1, the complete table would look as follows:

0	00000000	00000010	B B B 00001000	00000001	01001000	0	2	8	1	H
1	00000010	00000000	B B B 00001010	00000001	01010111	2	0	10	1	W
2	00000010	00000000	B B B 00000111	00000001	01101100	2	0	7	1	l
3	00000000	00000000	B B B 00000001	00000001	00100000	0	0	1	1	_
4	00000000	00000010	B B B 00000101	00000010	01101100	0	2	5	2	l
5	00000000	00000010	B B B 00000011	00000001	01101111	0	2	3	1	o
6	00000000	00000010	B B B 00000000	00000001	00100001	0	2	0	1	!
7	00000010	00000000	B B B 00000110	00000001	01100100	2	0	6	1	d
8	00000000	00000010	B B B 00000100	00000001	01100101	0	2	4	1	e
9	00000010	00000000	B B B 00000010	00000001	01110010	2	0	3	1	r
10	00000010	00000000	B B B 00001001	00000001	01101111	2	0	9	1	o

Here is the what the colours used here stand for:

0	BLACK
2	GREEN

And so, here is what the message is actually supposed to look like:

Hello World !

If foreground and background colour are the same you would end up with unreadable text. Luckily, the scientists have figured what to do when this happens! When these two values are the same, then they should be used to print special effects instead. The scientists have not figured all the values out yet, but we already know of the following:

0	reset to normal
26	stop blinking
42	bold
66	faint
105	conceal
153	reveal
182	blink

1. Copy over your solution from “decode.s”.
2. Implement your new `decode` subroutine. To make it easiest, first start with printing the colours correctly.
3. After colours have been properly implemented, now implement the printing of effects.

Note for WSL users: Support for these special effects depends on the terminal you are using. To get them to work properly, we recommend using “Windows Terminal”, which is available in the Microsoft Store: <https://www.microsoft.com/store/productId/9N0DX20HK701>. See also 2.1.2

4.1.2 Required Reading

1. For this assignment, you must be familiar with ANSI Escape Codes. Kindly check section 3.1 of the assembly manual.
2. In addition, information about lookup tables and switch statements can be found in section 3.2.

4.2 ASSIGNMENT 5: Valid Parentheses (250 points)

In this assignment you will implement the `check_validity` subroutine. It will process a file containing multiple sequences, treating each one as a separate input string, and check whether each sequence is correctly parenthesized. A valid parenthesization means that the most recent type of open bracket is the first one that gets closed, i.e., the closed brackets should always appear after their corresponding open pair. Your subroutine must return the number of sequences that were found to be invalid.

To exemplify, the following sequences “{}()” and “<[()]>” are correctly parenthesized because for every open bracket there is a closing bracket of the same type placed in the proper spot. An example of an invalid sequence is: “(<)>” because the round bracket is closed before the matching pair of the angled one, a correct sequence could have been “(<>)”.

Constraints

1. There will be at least 1 character in the input string.
2. Your program should recognize the following pairs of brackets: round (), square [], curly {}, and angle <>.
3. The “priority” of parentheses is not relevant, i.e. both [()] and ([]) are valid.
4. The sequence may contain any other ASCII characters, but your subroutine must ignore them. For example, “[<>]” and “[br<as>il]” are treated identically.

Exercise:

1. Download the files for this assignment from Brightspace.
The file you will be writing your code in is “`check_validity.s`”. It currently includes the sequences from “`basic.s`” (the `.include ‘basic.s’` line does this). If you want to change the input sequences, simply change this line to include the file you want.
2. Implement your `check_validity` subroutine. We recommend you to first implement a subroutine to check one single string.
3. Test your program thoroughly. Provided are test files “`basic.s`” and “`basicInvalid.s`”.

4.3 ASSIGNMENT 6: Simplified printf Function (50–500 points)

The `printf` subroutine is just a subroutine like any other. To prove this, you will write your own simplified version of `printf` in this assignment. A list of the theoretical concepts that you require to do this assignment can be found under 4.3.1.

For this assignment, you can choose how simplified your `printf` really is! For each feature that you implement, you get a certain amount of points. Below you can see the point distribution, and each feature will be explained in more detail throughout this section. **Be aware:** you can have at most 2 sign-offs for this assignment!

1	50 pts	Printing without C library, and escaping %%
2	100 pts	Printing other strings with %s
3	100 pts	Printing unsigned integers with %u
4	50 pts	Printing signed integers with %d
5	100 pts	Printing 8-bit floats with %f
6	100 pts	Support for more than 6 arguments

The most basic solution must be able to print a format string (this is your first argument). For this exercise, you are not allowed to use the `printf` function or any other C library function. This means you will have to use system calls for the actual printing. Your function must follow the proper x86-64 calling conventions. To warm you up for the format specifiers, this solution must also be able to escape the % sign: when “%%” is encountered in the string, your subroutine should only print “%”. Give your `printf` function the name `my_printf` to avoid confusion with the real `printf` function in the C library.

If you then choose to implement any of the other features, your `printf` must be able to receive a **variable** amount of arguments. The rest of the arguments are printed instead of the placeholders (also called format specifiers) in the format string. How those arguments are printed depends on the corresponding format specifiers. Your `printf` function has to support any number of format specifiers in the format string.

Unlike the real `printf`, your version only has to understand the format specifiers listed below. If a format specifier is not recognized, it should be printed without modification.

Supported format specifiers:

%% Print a percent sign. This format specifier takes no argument.

%s Print a null terminated string. No format specifiers should be parsed in this string. The corresponding parameter is the address of first character of the string.

%u Print an unsigned integer in decimal. The corresponding parameter is a 64-bit unsigned integer.

%d Print a signed integer in decimal. The corresponding parameter is a 64-bit signed integer.

%f Print a signed float in decimal. The corresponding parameter is an 8-bit floating point number. We use the course’s own interpretation of FP8, which we kindly named **CO-FP8E4M3FNAN**. This means you will have 1 sign bit, 4 exponent bits with a bias of 7, and 3 mantissa bits. The **FNAN** part means “Finite” and “Always Normalized”, which basically excludes all the weird behaviours that you may see when doing research. Therefore, you can simply do the typical math with mantissa and exponent.

Example:

Suppose you have the following format string:

My name is %s. I think I’ll get a %u for my exam. What does %r do? And %%?

Also suppose you have the additional arguments “Piet” and 10. Then your subroutine should output:

My name is Piet. I think I’ll get a 10 for my exam. What does %r do? And %?

Compiling your solution

To compile this solution, you are expected to use a different command due to the Submit feedback script works. Namely, instead of the usual command, you are expected to use:

```
as -no-pie -g -o printf.o yourfile.s && ld --fatal-warnings --entry main -o printf printf.o  
./printf
```

This command aligns the stack without a prologue being used to begin with. You will still need to write the prologue. However, you then need to ensure that you realign the stack.

Hints

To get started you may divide the work in a number of steps. Note that these are just hints, you do not have to follow these steps to finish this assignment.

1. Write a subroutine that prints a string using system calls.
2. Write a new subroutine to recognize format specifiers in the format string. Initially, you can discard the format specifiers rather than process them. The rest of the string can be printed using the subroutine from the previous hint.
3. Implement the various format specifiers. It may help to implement %u before %d. Again, you may use the print function from the first hint if you implemented it.
4. Since you are dealing with a variable amount of arguments, it may help to store all input argument registers on the stack at the start of your function, even if you don't end up using them.

4.3.1 Required Reading

1. In order to successfully complete this assignment, the information in section 3.3 of the assembly manual might be useful. This section elaborates on how I/O can be done without the C library.

4.4 ASSIGNMENT 7: Implement “diff” in Assembly (400 points)

For this exercise you will implement a simplified version of the Unix “diff” program in assembly. The purpose of the `diff` program is to compare two files line by line and show all the differences between them. A list of the theoretical concepts that you require to do this assignment can be found under 4.4.1.

Exercise

You can find a tarball containing more details for the implementation and assembly code for reading files specified as command line arguments in Brightspace. Implement a subroutine called `diff` that takes as input pointers to the two unprocessed files, a boolean for the `-i` flag, and a boolean for the `-b` flag. This subroutine should output the result directly to stdout in the format shown below. The subroutine must:

- Implement a line-by-line comparison version of `diff`: Compare lines in the two files in order and consider it a change if they differ in any way (there is no need to identify additions and deletions specifically). Line 1 of file A is compared to line 1 of file B, line 2 of file A is compared to line 2 of file B, and so on. Your `diff` must return 0 if no differences were found, or return 1 otherwise.
- Implement the `-i` and `-b` options that `diff` offers. Your program will take the name of two files to compare, and treat anything after that as options. They can be used in any order, and anything different from these two flags is considered an error. See the `diff` manual to learn what these options do (you can do so by running `man diff` in a terminal).

You must read the `-i` and `-b` options from the command line arguments. You will implement the `check_flags` subroutine to do so. By convention, you get your command line arguments the following way: first an integer that tells you how many arguments were provided (always at least 1; the name of your own program) and then the actual parameters. More information can be found online. This subroutine should return a number that allows us to properly call your `diff`. See the README and the comments above the subroutine in `diff.s` for more details.

Clarifications

Keep in mind that this assignment is a simplified version of `diff`. If anything is unclear, please ask for clarifications from a TA during the labs.

- Your program must handle files with different amounts of lines. When you have reached the end of the shorter file, you must keep printing changes until the other one was also entirely done. Follow the example shown below and print nothing as the content of the shorter file.
- Your output must follow the format shown below, with the exact contents of the printed lines as they were in the files, even when flags are used.
- You can treat the characters with ASCII codes 9, 11, 12, 13 and 32 as whitespaces. We use the Unix newline `'\n'`, which is different from the Windows newline `"\r\n"`, so you can treat the carriage return `'\r'` as any other whitespace character.
- Your solution is not allowed to have hardcoded `-i` and `-b` flags, and the comparison between the strings must be done manually by you. Not following either of these constraints will result in a fail for this assignment.

Example

As a sample input, consider the following two files:

1 Hi, this is a testfile.	1 Hi, this is a testfile.
2 Testfile 1 to be precise.	2 Testfile 2 to be precise.
3 The flags matter here.	3 The flags MATTER here.
4 The file ends here.	4 This line was added now.
	5 The file ends here.

The resulting output of your `diff`, with and without the flags, would then be:

```

$ ./diff file1.txt file2.txt
2c2
< Testfile 1 to be precise.
---
> Testfile 2 to be precise.
3c3
< The flags matter here.
---
> The      flags MATTER here.
4c4
< The file ends here.
---
> This line was added now.
5c5
<
---
> The file ends here.

```

```

$ ./diff file1.txt file2.txt -b -i
2c2
< Testfile 1 to be precise.
---
> Testfile 2 to be precise.
4c4
< The file ends here.
---
> This line was added now.
5c5
<
---
> The file ends here.

```

4.4.1 Required Reading

1. To further understand how the `diff` command works, please check the `diff` manual by running `man diff` in a terminal.
2. It might be useful to check section 3.2 of the assembly manual, which tackles the subject of switch statements and lookup tables.

4.5 ASSIGNMENT 8: Hide Data in a Bitmap (500 points)

A list of the theoretical concepts that you require to do this assignment can be found under 4.5.1.

The scientists from Assignment 3 (3.3.1) suggested that our generation should also leave a message for future generations. Frustrated at how long it took them to implement a decoder, the decision was made that this new effort should also be an “interesting” puzzle.

You are tasked by the scientists to encrypt messages using assembly, into everyday barcodes. This assembly code will then later be hacked into a popular barcode software package. In detail:

1. Make sure you understand the message, including the “lead” and “trail”.
2. Compress the message using the Run-Length Encoding (RLE) technique.
3. Prepare the barcode.
4. Use XOR to encrypt the message into the barcode.
5. Save the results as image bitmaps, in BMP format.
6. Test that you can decrypt the message, using again the XOR encryption technique.

Note : There should be a separate subroutine for encrypting the data. This takes a pointer to the input message, actually encrypts it and then returns a pointer to the bitmap data. Additionally, you should implement a decrypting subroutine which takes as input the bitmap bytes and returns a pointer to the original message (null terminated and without lead/trail).

Data: The Message

Complexity: Easy

To demonstrate that your assembly code works, encrypt and decrypt the message **The answer for exam question 42 is not F.**, including the final dot (‘.’). Save the message as an assembly data chunk before you continue.

Each message, before encryption, must be preceded and followed by the following pattern:

$$8 \times C, 4 \times S, 2 \times E, 4 \times 1, 4 \times 4, 8 \times 0$$

Here, $\langle \text{number} \rangle \times \langle \text{character} \rangle$ means that *character* is repeated *number* times, e.g., $8 \times C$ means CCCCCCCC. The added parts are called the “lead” and the “trail” of the message.

Data Compression: Run-Length Encoding (RLE)

Complexity: Moderate

After saving the message as a data chunk, the next step is to compress it. You will do so by using RLE-8. You have to devise your own algorithms for RLE-8-encoding and RLE-8-decoding the message described in the previous paragraph.

Data: Barcodes

Complexity: Easy

The message will be encoded in the first part of a barcode. This subsection details how the barcode can be created. For simplicity, imagine that the barcode looks like this:

W W W W W W W B B B B B B B W W W W B B B B W W B B B W W

This pattern is 31 pixels long. In this pattern, W stands for a white pixel and B stands for black. This pattern should always be followed by a red pixel.

Create a sequence by repeating this pattern, followed by a red pixel, 32 times. This will form a 32×32 image, where each pixel is either white, black, or red. Note that this key represents an RGB image, where each pixel is represented using three bytes.

Data Encryption: XOR

Complexity: Moderate

Now that the barcode is created, the RLE-encoded message can be encrypted. Your task is to encrypt the RLE-encoded message using the barcode image as key. Note that the key consists of $32 \times 32 \times 3$ bytes, while the message is much shorter. This means that only a small part of the image will change.

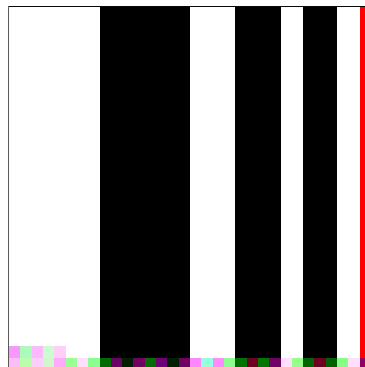
Data Representation: the BMP Format (Simplified)

Complexity: Difficult

Storing data as images requires complex data formats. One of the simplest is the bitmap (BMP) format, which you must use. BMP files encode raster images, that is, images whose unit of information is a pixel; raster images can be directly displayed on computer screens, as their pixel information can be mapped one-to-one to the pixels displayed on the screen. Therefore, your task is to save the encrypted message from step 4 in this data format.

Notes:

1. You are not allowed to use any other format. You can check the assembly manual for additional explanations on BMP.
2. As an example, the following image contains the message **The quick brown fox jumps over the lazy dog** encrypted in the barcode pattern. Note that the message will be quite more visible in this example than what one would do in reality.



Last But Not Least

Make sure that you are also able to decrypt the message. After this, you should go and have your code checked by a lab course assistant. You have now officially proved mastery in the basics of assembly programming. Not bad!

4.5.1 Required Reading

1. You can learn more about RLE in section 3.5 of the assembly manual.
2. For additional information about XOR, kindly check section 3.6 of the assembly manual.
3. In addition, section 3.7 of the assembly manual describes the bitmap format that you are expected to use.

4.6 ASSIGNMENT 9: Branch Prediction (200–600 points)

While it may not be immediately apparent in higher-level languages, every if statement, while loop, for loop, etc., is composed of *conditional jump statements* in Assembly. When a processor is executing code and wants to efficiently pipeline the instructions, conditional statements can heavily reduce performance, because the condition has to finish execution before the next instruction can enter the fetch stage. An efficient pipeline like the four-stage fetch-decode-execute-write taught in lectures has to *guess* what the next instruction will be in order to already process it rather than being bottle-necked by the previous conditional statement finishing execution. The next instruction depends on whether the conditional jump occurs or not. If the processor *guesses* wrong, the pipeline is stalled and the other instruction has to be fetched instead, adding several clock cycles and hindering performance. So how can the processor guess efficiently with high accuracy? That leads us to one of the most beautiful fields of optimization in computer architecture: **branch prediction**.

There are many branch prediction techniques from simple 1-bit counters to sophisticated multi-layer perceptrons. We encourage you to look into these yourself and choose a strategy that best suits your interests and your performance goal for this assignment. You are **allowed and encouraged to use LLMs** to find out more about branch prediction algorithms and guide you. However, you must use LLMs responsibly; use AI to help explain new concepts and various branch predictor types, but **do not use it to generate code!** You will be assessed in the end based on your understanding of your own code, so while you may use LLMs to guide you, you must have a clear understanding of your submission, branch prediction in general, and its usefulness in computer architecture. This assignment is graded depending on the performance achieved (up to 600 points).

4.6.1 Implementation

You will be implementing a live, emulated branch predictor on real code such as Towers of Hanoi and rendering the Mandelbrot Set. These programs were written in a minimalist imperative language called brainfuck and will be run on an interpreter that is present in the folder with all the materials relevant for the assignment. This brainfuck interpreter¹ will call your `predict_branch` subroutine located in `branch_predictor.s` every time it needs to decide whether to enter or exit a loop.

You will complete the subroutine `predict_branch` which takes in the branch address as input in `%rdi` and returns whether to **not take** the branch (0) or to **take** the branch (1) in `%rax`.

Currently, the solution below implements a **static predictor**, i.e. it always takes the branch. At minimum, you must submit a **dynamic predictor**. For more points and better performance, you may delve into more advanced dynamic predictors.

```
# Arguments:
# %RDI: Branch address
# Return:
# %RAX: 0 for not taken, 1 for taken (your prediction)
predict_branch:
    pushq    %rbp
    movq     %rsp, %rbp

    # Your predictor goes here
    movq     $1, %rax # Dummy predictor: always take branch

    movq     %rbp, %rsp
    popq     %rbp
    ret
```

To inform the predictor of actual taken branches such that it can learn (mandatory for any dynamic predictor), the following `actual_branch` subroutine is called after every outcome of a conditional is already known (regardless of what was predicted). Here, implement the subroutine such that you store the information of past actual outcomes. Refer to the hints for more guidance!

¹The implementation is a modified version of Oleg Mazonka's very fast brainfuck interpreter

```

# Arguments:
# %RDI: Branch address
# %RSI: Actual outcome
actual_branch:
    pushq    %rbp
    movq     %rsp, %rbp

    # Store the actual outcome to train the predictor

    movq     %rbp, %rsp
    popq     %rbp
    ret

```

In addition to these two, a `init` subroutine is present in the framework for the solution. It will be called once before the testing suite is run and may prove useful in more advanced implementation to initialize memory.

```

init:
    pushq    %rbp
    movq     %rsp, %rbp

    # (Optional) Initialize potential history tables

    movq     %rbp, %rsp
    popq     %rbp
    ret

```

4.6.2 Testing your solution

To test your submissions, you can download the Brightspace material and run the following command to test your predictor on a simple program.

```
make test          # compiles an example test
```

Note that passing the test example does not mean that you will be able to get a successful submission. The tests we will run your program against are divided into two categories: once you pass more than 80% of the easy tests you will receive points for this assignment and your program will be run on some more difficult tests.

Specifically, you receive $200 + \frac{\text{percentage of medium test passed} - 50}{50} \cdot 400$. This can get you to a maximum of 600 points for the assignment (although it is almost impossible to get more than 599 points due to the learning phase of the branch predictor).

4.6.3 Hints

1. Branch predictions rely on learning patterns based on **past predictions** to predict future outcomes. Therefore, it would be wise to store some sort of history table.
2. Every branch is defined by its instruction pointer, passed in `%rdi`. You can be certain that this value uniquely represents the single branch and will not change over time.
3. You can use the branch address (instruction pointer) as a hash value in some map that you are storing the history in.
4. You can assume that there will be at most 2048 unique branches per execution of the predictor.
5. You can assume that the branch pointer will be at most 64 bits long and different from `0x00`.

Bonus: If you want to read more about a puzzling result of branch prediction when looping over a sorted vs. unsorted list with one of the most upvoted posts on StackOverflow, refer to this post. Try thinking about why this happens yourself before looking at the answers!

4.7 ASSIGNMENT 10: Brainfuck (400 points)

For the assignment description, check 4.7.1. For a link of possible programs to test the speed of your solution on, check 4.7.2.

4.7.1 Assignment description

For this assignment, you will implement an interpreter for the very simple “programming language” called Brainfuck.² In Brainfuck, every character of the source code is a command, and there are only eight very simple commands. The interesting thing about this language is that even though it might not seem like it, it is still Turing complete. (This basically means that you can write every possible program in it, but not necessarily in an efficient way.)

Exercise:

You can find a tarball containing assembly code and instructions for reading a file specified as a command line argument on Brightspace. You should write a **brainfuck** subroutine that takes a pointer to the Brainfuck code as argument and executes this program.

Alternatively, if you want to go above and beyond, you can also try to implement a Brainfuck compiler! There will be a different assignment for submitting compilers (see why after the examples below). You can only get points for this assignment once, so even if you submit both an interpreter and a compiler, you will only get 400 points.

Examples:

hello.b:

```
>+++++++[<+++++++>-]<.>+++++++[<++++>-]<+.+++++. .+++.>>+++++++
[<++++>-]<.>>>+++++++[<+++++++>-]<---. <<<<.+++ .----- .----- .>>+.
```

Now, executing your program as follows

```
./brainfuck hello.b
```

should make it give

```
Hello World!
```

as output.

cat.b:

```
,[.,]
```

This program is the equivalent of the Linux command **cat**; running this program will copy whatever you enter in the console. By sending the null character to the console, the loop halts (this is done by pressing **Ctrl+2** in the terminal).

Bonus points:

The larger the Brainfuck program, the slower your interpreter might get. At some point, it might feel even tedious to wait for the result! Can you make your interpreter faster?

Find ways to significantly improve the speed or memory usage of your interpreter. At the end of this course, we will run the latest submission of groups that had a sign-off and see which ones are the fastest. To keep things fair, there will be two separate competitions, between interpreters and between compilers. If you submit a solution to both competitions, you can only get bonus points for one of them.

Everybody who submits and signs off a valid interpreter/compiler will receive 400 points.

4.7.2 Required Reading

1. We encourage you to try more complex programs to test your interpreter. For instance, we recommend <http://esoteric.sange.fi/brainfuck/utils/mandelbrot/mandelbrot.b>.

²<https://esolangs.org/wiki/Brainfuck>

4.8 ASSIGNMENT 11: Implement a Game (750 points)

This section details the last bonus assignment of the course. For the description, refer to 4.8.1. For some hints, kindly check 4.8.2

4.8.1 Assignment description

For this exercise you will use your basic assembler programming skills to develop a complete and useful program: a game. For the purpose of this assignment, a game is an interactive computer application in which at least one user (player) influences the outcome via keyboard or mouse input.

You are free to choose any game, except for simple text-based games such as “Guess a number” or Trivia. The game you choose does not have to be overly complex either; we suggest implementing a (single-player) Pong game, where the player controls a paddle and tries to prevent the ball from crossing the player’s goal line, or something of comparable difficulty.

Before you start programming, you must hand in a specification (about one A4, PDF) of your game via Submit (Bonus 8 Specification) and send an email to co-cs-ewi@tudelft.nl (subject ”A8 Bonus Specification”). We will check whether your idea is reasonable and sufficient. We will reply to your email with details and approve your specification or request changes.

Specification Requirements

A reasonable specification should clarify the following aspects:

1. Whether you are implementing the game alone or with one more person.
2. What OS are you planning to use? Will it be the one provided by us?
3. What libraries do you plan on using, if any?
4. How do you plan on rendering the game graphics?
5. Even in the case of a common game, do briefly explain the rules that you are going to implement. This ranges from aspects such as:
 - (a) User input. Will it be mouse-based, keyboard-based, or both? Will it be something else?
 - (b) When does the game end? What are the win/lose/draw conditions, if any?

Implementation Requirements

Your task is to implement a game, subject to the following requirements:

1. Your implementation should correctly implement the rules of the game.
2. Your implementation should display the state of the game. If the rules of the game make it possible, your implementation should display the current progress of the player toward achieving the goal of the game.
3. Your implementation should permanently record the top scores, if the rules of the game allow it. Your implementation should also have an option to display them. A bootable game only needs to store top scores until the next reboot (i.e., they do not need to be written to disk).
4. Your implementation should be able to receive input from at least one player. The input has to come either from the keyboard or from the mouse.

Important: For this extra assignment you should NOT expect content-related help from the lab assistants, that is, you should not expect the lab assistants to debug your code or even to suggest how you should solve the assignment. Instead, this assignment allows you to demonstrate your ability to go beyond the manual (and lab assistant).

4.8.2 Hints about Solving this Assignment

Hint #1: Although we are sure you know how to use a search engine and understand the keywords you need to search for this assignment, we would like to give you a hint for writing games in assembly. For the past 15 years, the demo scene and several generations of students around the world have learned much from Denthor's Asphyxia Tutorials, <http://archive.gamedev.net/archive/reference/listed82.html?categoryid=130>. In particular, you may be interested in graphics (tutorial Mode 13h, <http://archive.gamedev.net/archive/reference/articles/article347.html>).

Hint #2: Bootable game Another option is to make your game bootable, so that it can run without any operating system. This means that you will have access to the graphical (VGA) memory directly, implement interrupt handlers to work with timing and input, and basically be able to (but also have to) do everything yourself. There are no standard libraries either. (So no printf, no exit, no nothing.) These advantages may also be disadvantages.

You do not need to start from scratch in implementing a bootable game. You can use a simple OS/library that we made and that will set up most of the basic things for you. This library switches the processor to 64 bit mode, makes sure you have a stack, and provides an API for you. After that, you are basically on your own. Again, many advantages, but these can turn easily into disadvantages.

This basic OS can be found on <https://gitlab.ewi.tudelft.nl/computer-organization/lab-related/gamelib-x64>.

Last But Not Least

You should go and have your code checked by a lab course assistant. You have now officially proved excellence in mastering the basics of assembly programming. Congratulations, you have completed the most difficult part of the lab!